

Spectral Gap Bounds for Random-Walk Metropolis Samplers on Log-Concave Targets

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Abstract

We derive sharpened upper and lower bounds on the spectral gap of the Random-Walk Metropolis (RWM) algorithm targeting a strongly log-concave density on $\mathbb{R}^d$. Building on the conductance arguments of Lovasz and Simonovits and the canonical-path machinery of Diaconis and Stroock, we show that the optimal Gaussian proposal variance scales as $\sigma^2 = \Theta(d^{-1})$ and that the resulting gap is bounded below by c/Ld^{-1} for a strong-convexity constant $L > 0$ and a universal constant $c \in (0, 1)$. A complementary upper bound, obtained by exhibiting an explicit slow-mixing test function, shows the rate is tight up to a factor logarithmic in the condition number. Numerical experiments on a battery of three benchmark targets — the isotropic Gaussian, a moderately ill-conditioned anisotropic Gaussian, and a Bayesian logistic-regression posterior — corroborate the theory and expose the role of preconditioning. We close with implications for adaptive samplers, in which the proposal covariance is learned on the fly from the chain’s empirical second moment.

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1. Introduction

Markov chain Monte Carlo (MCMC) is the workhorse of modern Bayesian computation. Given a target density π on $\mathbb{R}^d$ known only up to its normalising constant, MCMC algorithms construct a Markov chain whose stationary distribution coincides with π and whose ergodic averages approximate posterior expectations. Among such algorithms the Random-Walk Metropolis (RWM) sampler of Metropolis et al. [1] and its symmetric-proposal generalisation due to Hastings [2] remain the simplest and most widely deployed: their only tuning parameter is the proposal step size, and they require nothing more than pointwise evaluation of $\log \pi$.

The Achilles' heel of RWM is its $O(d)$ slowdown as the dimension of the target grows — a phenomenon first quantified by Roberts, Gelman, and Gilks [3] for product targets and subsequently extended to a broader class of log-concave densities by Bedard [4] and by

Beskos, Roberts, and Stuart [5]. These optimal-scaling results identify $\sigma^2 = O(d^{-1})$ as the right asymptotic regime for the proposal variance and pin the optimal expected acceptance rate at the celebrated value 0.234. What they leave open is a tight bound on the spectral gap¹ — the quantity controlling the relaxation time of the chain.

P and stationary distribution π is $1 - \lambda_2$, where λ_2 is the second-largest eigenvalue of P acting on $L^2(\pi)$. Larger is better: a gap of g implies relaxation in $O(g^{-1})$ steps from any $L^2(\pi)$ -bounded initial distribution.

In this paper we close the gap (no pun intended) by combining two classical tools: the *conductance* framework of Lovasz and Simonovits [6] — the Markov-chain analogue of Cheeger's inequality — and the *canonical-paths* technique of Diaconis and Stroock [7]. For a strongly log-concave target with strong-convexity constant L , we prove (Theorem @thm:gap below) that the gap of RWM with optimally-tuned step size is at least $c L / d$ for a universal constant $c > 0$; the upper bound, established in section 3, matches this rate up to logarithmic factors in the condition number of the target's Hessian.

The remainder of the paper is structured as follows. Section 2 fixes notation and states the assumptions. Section 3 develops the lower and upper spectral-gap bounds and gives a worked proof of the lower bound. Section 4 reports numerical experiments on three benchmark targets, including a figure summarising the autocorrelation decay (Figure ?) and tables documenting the empirical effective sample size (Tables 2.1 and 3.1). Section 5 discusses the implications for adaptive samplers, and section 6 collects references.

2. Method

Throughout the paper π denotes a probability density on $\mathbb{R}^d$ with respect to Lebesgue measure, normalised so that $\int \pi(x), dx = 1$. We write $U(x) = -\log \pi(x) + \text{const}$ for the associated potential and assume $U \in C^2(\mathbb{R}^d)$ with

¹The spectral gap of a reversible chain with transition kernel

Hessian satisfying

$$L\mathbb{I}_d \preceq \nabla^2 U(x) \preceq M\mathbb{I}_d, \quad \forall x \in \mathbb{R}^d,$$

where $0 < L \leq M < \infty$ are the strong-convexity and smoothness constants and $\mathbb{I}_d$ is the $d \times d$ identity. The *condition number* of the target is $\kappa = M/L$; small κ means a well-rounded target, large κ a pencil-shaped one.

The Random-Walk Metropolis chain with isotropic Gaussian proposal is defined by the transition kernel

$$P(x, A) = \int \alpha(x, y) q(y|x) dy + \delta_A(x), \quad \text{where } \alpha(x, y) = \min\left\{1, \frac{q(x|y)\pi(y)}{q(y|x)\pi(x)}\right\}.$$

where $q(y|x)$ is the proposal density and $\alpha(x, y) = \min\{1, \pi(y)/\pi(x)\}$ is the Metropolis acceptance probability. Reversibility with respect to π is immediate from the symmetry of the proposal q .

The Metropolis acceptance probability $\alpha(x, y) = \min\{1, \pi(y)/\pi(x)\}$ always yields a π -reversible chain. This is one of the principal practical advantages of RWM over alternatives such as Hamiltonian Monte Carlo, which require a more delicate symplectic integrator to preserve detailed balance.

2.1. Conductance and the Cheeger inequality

Recall that the *conductance* of a π -reversible Markov chain P is

$$\Phi(P) = \inf_{0 < \pi(S) \leq 1/2} \frac{\int_S P(x, S^c) \pi(dx)}{\pi(S)}.$$

The discrete Cheeger inequality [8] then bounds the spectral gap by

$$\frac{\Phi^2}{2} \leq \text{gap}(P) \leq 2\Phi.$$

The strategy of section 3 is to lower-bound Φ directly, via an isoperimetric inequality for log-concave measures, and then upper-bound $\text{gap}(P)$ via an explicit slow-mixing test function.

2.2. Canonical paths

The complementary canonical-paths machinery of Diaconis and Stroock [7] expresses the gap in terms of the maximum congestion along a system of paths connecting every pair of states. For our continuous-state RWM chain we discretise via the standard lattice covering of the target's effective support and apply the

canonical-paths bound to the resulting random walk on a graph. The full derivation, which we will not reproduce here, follows the template laid out by Vempala [9] for hit-and-run.

2.3. Test targets

We verify the theory on three target families summarised in Table 2.1. The isotropic Gaussian is the textbook well-conditioned target ($\kappa = 1$); the anisotropic Gaussian probes the effect of moderate ill-conditioning; the logistic-regression posterior is a realistic non-Gaussian target with both curvature variation and a high-dimensional parameter space.

Target	d	κ	log-concave?
Isotropic Gaussian	32	1	yes
Anisotropic Gaussian	32	10^2	yes
Bayesian logistic posterior	64	≈ 8	yes

Table 2.1.

For each target we run RWM at the asymptotically optimal step size $\sigma^2 = 2.38^2/d$ recommended by Roberts et al. [3] and record the integrated autocorrelation time and the effective sample size per second of wall clock.

3. Results

3.1. Spectral-gap lower bound

Our principal theoretical contribution is the following.

Theorem 1: Let π be a probability density on $\mathbb{R}^d$ whose potential $U = -\log \pi$ satisfies $L \mathbb{I} \leq \text{grad}^2 U \leq M \mathbb{I}$ pointwise, with $0 < L \leq M < \infty$. Then the Random-Walk Metropolis chain with proposal variance $\sigma^2 = c_1$ and proposal covariance equal to a scalar multiple of the identity satisfies

$$\text{gap}_d(P) \geq c_2 L$$

where c_1 and c_2 are universal constants depending only on the acceptance rate, not on d, L , or M .

Proof: By Cheeger's inequality it suffices to show $\Phi(P) \geq c_3 \sqrt{L/d}$ for some universal $c_3 > 0$. For any measurable S with $\pi(S) \leq 1/2$, the numerator integral $\int_S P(x, S^c) \pi(dx)$ is bounded below by the probabil-

²A symmetric proposal $q(x, y) = q(y, x)$ together with

ity that an $N(0, \sigma^2 I)$ -distributed proposal crosses the boundary of S and is then accepted. Strong log-concavity ensures that the chord-length distribution from any interior point to the boundary of S is sub-exponential with rate $\sqrt{L}$; standard Gaussian small-ball estimates then give the crossing probability as $\Omega(\sigma \sqrt{L})$. Choosing $\sigma = \sqrt{c/d}$ yields $\Phi \geq c^{3/2} \sqrt{L/d}$. The lower bound on the gap then follows from $\frac{\text{gap}}{2} \geq \frac{\Phi}{2} \geq \frac{c^{3/2}}{2} \sqrt{\frac{L}{d}}$, as required.

[#thm:gap]

The constants in the proof above are not optimal; tightening them requires a more delicate use of the Bakry-Emery criterion and is left to future work.

3.2. Spectral-gap upper bound

For the upper bound we exhibit the explicit test function $f(x) = x_1 - \pi[X_1]$ (the first coordinate, centred) and compute the Dirichlet form $\mathcal{E}(f, f)$. A short calculation shows

$$\mathcal{E}(f, f) \leq \frac{C}{L} \text{Var}_\pi(f),$$

which by the variational characterisation of the spectral gap implies $\text{gap}(P) \leq C L/d$ up to a $\log \kappa$ factor. Combined with Theorem @thm:gap, this pins the gap at $\Theta(L/d)$ in the well-conditioned case.

3.3. Empirical effective sample size

Table 3.1 reports effective-sample-size-per-iteration (ESS/n) estimates from $n = 10^6$ post-burn-in samples on each of the three benchmark targets, averaged over 32 independent chains.

Target	σ^2	Acceptance	ESS/n	τ_{int}
Isotropic Gaussian	$2.38^2/d$	0.236	1.34×10^{-2}	37.3
Anisotropic	$2.38^2/d$	0.218	2.05×10^{-3}	244
Logistic posterior	$2.38^2/d$	0.231	5.81×10^{-3}	86.1

Table 3.1.

The isotropic-Gaussian column matches the theoretical rate $\text{gap} = \Theta(d^{-1})$ to within a factor of two; the anisotropic column shows the κ^{-1} penalty predicted by the upper bound; the logistic posterior sits between the two, consistent with its modest effective condition number.

[SVG file [trapezoid_convergence.svg](#) omitted in non-SVG back-end]

Figure ?.

Empirical autocorrelation of the first coordinate for the three benchmark targets at $n = 10^6$ samples each. The isotropic target relaxes within ≈ 40 steps; the anisotropic target requires ≈ 250 ; the logistic posterior is intermediate. Data drawn from Table 3.1.

The autocorrelation curves in Figure ? decay at rates consistent with the spectral gaps derived in Theorem @thm:gap. Empirically the slope of the log-autocorrelation plot tracks L/d to within a factor of three across all three targets, suggesting that the universal constant c_2 of Theorem @thm:gap is closer to $1/3$ than to $1/100$.

3.4. A diagnostic: trace-plot stationarity

For each of the three chains we monitored the cumulative running mean $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ of the first coordinate and the Gelman-Rubin potential-scale-reduction factor $\widehat{R}$ across 32 independent replicates. All chains achieved $\widehat{R} < 1.01$ within the burn-in window of 5×10^4 iterations; the anisotropic target was the slowest to converge by this diagnostic, requiring $\approx 3 \times 10^4$ iterations against the isotropic target's $\approx 5 \times 10^3$ — a factor-of-six gap that again matches the κ^{-1} penalty predicted by the upper bound.

3.5. Robustness to the proposal-variance choice

To confirm that the asymptotically optimal scaling $\sigma^2 \propto d^{-1}$ is not an artefact of our particular constant of proportionality, we re-ran the isotropic Gaussian experiment with $\sigma^2 = c \times 2.38^2/d$ for $c \in \{0.25, 0.5, 1, 2, 4\}$. The resulting ESS/n varied by no more than a factor of two across the five settings, with the maximum at $c = 1$, in agreement with the prediction of Roberts et al. [3]. This wide plateau is one of the redeeming features of RWM in practice: the chain is forgiving of a step-size misspecification of up to a factor of two in either direction, which makes hand-tuning realistic in moderate-dimensional applications.

4. Discussion

The analysis above sharpens three folk theorems and exposes one less familiar subtlety.

First, the celebrated 0.234 optimal acceptance rate [3] is *not* an artefact of product targets: our The-

orem [@thm:gap](#) holds for any strongly log-concave target and recovers the same asymptotic rate $\sigma^2 = \Theta(d^{-1})$ as the right scaling for the proposal step size. The universal constant c_2 does depend, weakly, on the chosen acceptance rate, but the dependence is non-monotone and shallow — our Theorem holds for any acceptance rate bounded away from 0 and 1.

Second, *preconditioning matters*. The naive isotropic-proposal RWM pays a factor κ in mixing time for ill-conditioned targets; the preconditioned variant in which the proposal covariance is set to the inverse Hessian of U at the mode (or, more practically, to a running estimate of the chain’s empirical covariance) recovers the well-conditioned rate. This is the principal practical justification for adaptive samplers such as the Haario-Saksman-Tamminen adaptive Metropolis [10] and the more recent Metropolis-adjusted Langevin algorithm with preconditioning by the empirical Fisher information [11].

Third, the *gap-vs.-mixing-time gap* (the factor $\log(1/\epsilon)$ that appears when converting a spectral-gap bound into a total-variation mixing-time bound) is well controlled in our setting. Strong log-concavity of the target implies a logarithmic Sobolev inequality with constant L , which in turn upgrades the spectral-gap bound to a TV-mixing-time bound of $\tilde{O}(d/L)$ steps for any target accuracy ϵ in $(0, 1)$.

The subtler observation, mentioned only in passing in section 4, is that the *anisotropy penalty* for RWM is sharper than for Hamiltonian Monte Carlo or for Langevin-based samplers. Where RWM pays a factor κ in the mixing time, HMC with a suitably-tuned integrator pays only $\sqrt{\kappa}$, and Langevin-based samplers pay a factor that depends on the integrator’s order. Quantifying this trade-off in the spectral-gap language remains open; we view it as the most interesting avenue for follow-up work.

4.1. Implications for software libraries

Three concrete recommendations follow for practitioners.

1. **Use an adaptive preconditioner.** The Haario-Saksman-Tamminen recipe [10] of replacing the isotropic proposal covariance with a running estimate of the empirical chain covariance recovers the κ penalty essentially for free, at the cost of a modest bookkeeping burden. All mainstream MCMC libraries (Stan, PyMC, Turing.jl, BUGS) already implement this in some form.
2. **Monitor the spectral gap, not the acceptance rate.** While the 0.234 acceptance rate is a useful

indicator of *near-optimal* tuning, it is a one-dimensional summary of a multi-dimensional object. An adaptive sampler that targets a fixed acceptance rate can be fooled by anisotropic targets into accepting too eagerly in the well-conditioned directions and too rarely in the ill-conditioned ones. The integrated autocorrelation time τ_{int} is a more robust monitor.

3. **Prefer Hamiltonian or Langevin variants for $d \gtrsim 50$.** The $\sqrt{\kappa}$ -vs.- κ scaling advantage compounds in high dimensions; for $d \geq 50$ the cost of the gradient evaluations required by HMC or MALA is usually amortised within a single relaxation time.

5. Conclusion

We have provided matching spectral-gap upper and lower bounds for the Random-Walk Metropolis algorithm targeting a strongly log-concave density on $\mathbb{R}^d$. The bounds match in the well-conditioned case ($\kappa = O(1)$) and disagree by at most a $\log \kappa$ factor in the ill-conditioned case; empirical evidence on the three benchmark targets (Table 3.1, Figure ?) corroborates the rates to within small constants. A natural follow-up would be a parallel analysis of preconditioned variants of RWM, in which the proposal covariance is learned adaptively from the chain’s empirical moments[3]¹ — a setting in which the spectral-gap techniques applied here will require modification to handle the loss of the Markov property.

because the proposal at step n depends on the entire history $X_0, \dots, X_{n-1}$. Ergodicity is then restored via a diminishing-adaptation condition. The relevant theory is laid out in Roberts and Rosenthal’s review [12].

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